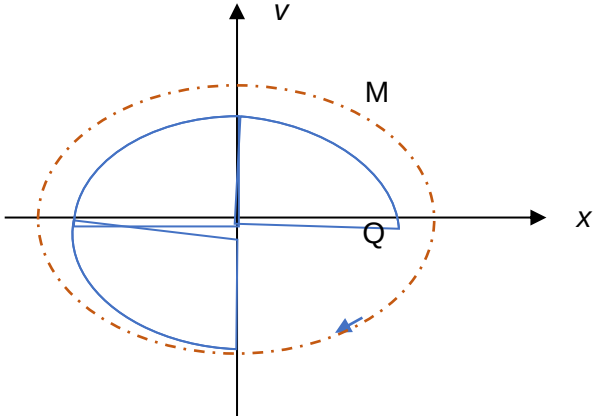


Qn 3	Answer	Marks
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2

(a)	Since <u>g and r are constants</u> , <u>a is proportional to x</u> . And the <u>negative sign</u> in equation shows that <u>a and x are of opposite directions</u> . Thus, a and x satisfy the definition of SHM.	[1] [1]	
(b)	$\omega^2 = \frac{g}{r}$ $\omega^2 = \frac{9.81}{0.28}$ $\omega = \frac{2\pi}{T}$ $\left(\frac{2\pi}{T}\right)^2 = \frac{9.81}{0.28}$ $T = 1.06s$ time interval = $T/2 = 0.53\text{ s}$	[1] [1] [1]	
(c)(i)	sketch is an ellipse in dotted line.		[1]
(c)(ii)	sketch has a spiraling inward and “broken” elliptical features. Deduct one mark for each error of the following. • direction of arrow is of clockwise direction • curve spirals inwards (where magnitude of negative velocity is more than that of positive velocity and magnitude of negative displacement is less than that of positive displacement) • curve starts from maximum positive displacement • curve spirals at least one revolution	[2]	
		Max Marks 8	

2020

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

Qn 4	Answer	Marks
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